Service levels and allocated capacity

Cloud Manager 3.8

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Service levels and allocated capacity

The cost for Cloud Volumes Service for AWS is based on the *service level* and the *allocated capacity* that you select. Selecting the appropriate service level and capacity helps you meet your storage needs at the lowest cost.

Considerations

Storage needs include two fundamental aspects:

- The storage *capacity* for holding data
- The storage *bandwidth* for interacting with data

If you consume more storage space than the capacity you selected for the volume, the following considerations apply:

- You will be billed for the additional storage capacity that you consume at the price defined by your service level.
- The amount of storage bandwidth available to the volume does not increase until you increase the allocated capacity size or change the service level.

Service levels

Cloud Volumes Service for AWS supports three service levels. You specify your service level when you create or modify the volume.

The service levels are catered to different storage capacity and storage bandwidth needs:

- **Standard** (capacity)

  If you want capacity at the lowest cost, and your bandwidth needs are limited, then the Standard service level might be most appropriate for you. An example is using the volume as a backup target.

  - Bandwidth: 16 KB of bandwidth per GB provisioned capacity
- **Premium** (a balance of capacity and performance)

  If your application has a balanced need for storage capacity and bandwidth, then the Premium service level might be most appropriate for you. This level is less expensive per MB/s than the Standard service level, and it is also less expensive per GB of storage capacity than the Extreme service level.

  - Bandwidth: 64 KB of bandwidth per GB provisioned capacity
- **Extreme** (performance)

  The Extreme service level is least expensive in terms of storage bandwidth. If your application demands storage bandwidth without the associated demand for lots of storage capacity, then the Extreme service level might be most appropriate for you.

  - Bandwidth: 128 KB of bandwidth per GB provisioned capacity

Allocated capacity

You specify your allocated capacity for the volume when you create or modify the volume.

While you would select your service level based on your general, high-level business needs, you should select your allocated capacity size based on the specific needs of applications, for example:

- How much storage space the applications need
- How much storage bandwidth per second the applications or the users require

Allocated capacity is specified in GBs. A volume's allocated capacity can be set within the range of 100 GB to 100,000 GB (equivalent to 100 TBs).

Number of inodes

Volumes less than or equal to 1 TB can use up to 20 million inodes. The number of inodes increase by 20 million for each TB you allocate, up to a maximum of 100 million inodes.

- <= 1TB = 20 million inodes
- >1 TB to 2 TB = 40 million inodes
- >2 TB to 3 TB = 60 million inodes
- >3 TB to 4 TB = 80 million inodes
- >4 TB to 100 TB = 100 million inodes

Bandwidth

The combination of both the service level and the allocated capacity you select determines the maximum bandwidth for the volume.

If your applications or users need more bandwidth than your selections, you can change the service level or increase the allocated capacity. The changes do not disrupt data access.

Selecting the service level and the allocated capacity

To select the most appropriate service level and allocated capacity for your needs, you need to know how much capacity and bandwidth you require at the peak or the edge.

List of service levels and allocated capacity

The leftmost column indicates the capacity, and the other columns define the MB/s available at each capacity point based on service level.

See Contract subscription pricing and Metered subscription pricing for complete details on pricing.

Capacity (TB)	Standard (MB/s)	Premium (MB/s)	Extreme (MB/s)
0.1 (100 GB)	1.6	6.4	12.8
1	16	64	128
2	32	128	256

Capacity (TB)	Standard (MB/s)	Premium (MB/s)	Extreme (MB/s)
3	48	192	384
4	64	256	512
5	80	320	640
6	96	384	768
7	112	448	896
8	128	512	1,024
9	144	576	1,152
10	160	640	1,280
11	176	704	1,408
12	192	768	1,536
13	208	832	1,664
14	224	896	1,792
15	240	960	1,920
16	256	1,024	2,048
17	272	1,088	2,176
18	288	1,152	2,304
19	304	1,216	2,432
20	320	1,280	2,560
21	336	1,344	2,688
22	352	1,408	2,816
23	368	1,472	2,944
24	384	1,536	3,072
25	400	1,600	3,200
26	416	1,664	3,328
27	432	1,728	3,456
28	448	1,792	3,584
29	464	1,856	3,712
30	480	1,920	3,840
31	496	1,984	3,968
32	512	2,048	4,096
33	528	2,112	4,224
34	544	2,176	4,352
35	560	2,240	4,480

Capacity (TB)	Standard (MB/s)	Premium (MB/s)	Extreme (MB/s)
36	576	2,304	4,500
37	592	2,368	4,500
38	608	2,432	4,500
39	624	2,496	4,500
40	640	2,560	4,500
41	656	2,624	4,500
42	672	2,688	4,500
43	688	2,752	4,500
44	704	2,816	4,500
45	720	2,880	4,500
46	736	2,944	4,500
47	752	3,008	4,500
48	768	3,072	4,500
49	784	3,136	4,500
50	800	3,200	4,500
51	816	3,264	4,500
52	832	3,328	4,500
53	848	3,392	4,500
54	864	3,456	4,500
55	880	3,520	4,500
56	896	3,584	4,500
57	912	3,648	4,500
58	928	3,712	4,500
59	944	3,776	4,500
60	960	3,840	4,500
61	976	3,904	4,500
62	992	3,968	4,500
63	1,008	4,032	4,500
64	1,024	4,096	4,500
65	1,040	4,160	4,500
66	1,056	4,224	4,500
67	1,072	4,288	4,500
68	1,088	4,352	4,500

Capacity (TB)	Standard (MB/s)	Premium (MB/s)	Extreme (MB/s)
69	1,104	4,416	4,500
70	1,120	4,480	4,500
71	1,136	4,500	4,500
72	1,152	4,500	4,500
73	1,168	4,500	4,500
74	1,184	4,500	4,500
75	1,200	4,500	4,500
76	1,216	4,500	4,500
77	1,232	4,500	4,500
78	1,248	4,500	4,500
79	1,264	4,500	4,500
80	1,280	4,500	4,500
81	1,296	4,500	4,500
82	1,312	4,500	4,500
83	1,328	4,500	4,500
84	1,344	4,500	4,500
85	1,360	4,500	4,500
86	1,376	4,500	4,500
87	1,392	4,500	4,500
88	1,408	4,500	4,500
89	1,424	4,500	4,500
90	1,440	4,500	4,500
91	1,456	4,500	4,500
92	1,472	4,500	4,500
93	1,488	4,500	4,500
94	1,504	4,500	4,500
95	1,520	4,500	4,500
96	1,536	4,500	4,500
97	1,552	4,500	4,500
98	1,568	4,500	4,500
99	1,584	4,500	4,500
100	1,600	4,500	4,500

Example 1

For example, your application requires 25 TB of capacity and 100 MB/s of bandwidth. At 25 TB of capacity, the Standard service level would provide 400 MB/s of bandwidth at a cost of $2,500 (estimate: see current pricing), making Standard the most suitable service level in this case.

Example 2

For example, your application requires 12 TB of capacity and 800 MB/s of peak bandwidth. Although the Extreme service level can meet the demands of the application at the 12 TB mark, it is more cost-effective (estimate: see current pricing) to select 13 TB at the Premium service level.

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